

**Department of Mechanical Engineering, Punjabi University, Patiala**  
**MCE -151 Engineering Graphics, First Mid-Semester Exam, Dec-2020**  
**B. Tech. 1st Year, 1<sup>st</sup> Sem (Group B)**

**Time: 1 Hour**

**Max. Marks: 15**

**NOTE: Section A is compulsory. Attempt any two questions from Section B.**

| Section A                                                        |                                                                                                                                                                                                                                    |             |
|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Q.1</b>                                                       | <b><u>All five parts of this question are compulsory</u></b>                                                                                                                                                                       | <b>(05)</b> |
| (i)                                                              | In oblique projection, the projectors drawn from the object are _____ to each other and _____ to the plane of projection. (Parallel, perpendicular, Inclined)                                                                      | 01          |
| (ii)                                                             | In third angle projection, the plane of projection lies between the _____ and _____.                                                                                                                                               | 01          |
| (iii)                                                            | A line is parallel to HP and inclined to VP, its front view reveals its _____. (Point view / true length / apparent length).                                                                                                       | 01          |
| (iv)                                                             | To obtain the point view of a line, the auxiliary plane must be drawn _____ to the true length of the line. (Parallel, Perpendicular).                                                                                             | 01          |
| (v)                                                              | A pentagonal lamina is inclined to HP and perpendicular to VP. Its front view will appear as _____. (Edge view / Apparent view / True Shape)                                                                                       | 01          |
| Section B                                                        |                                                                                                                                                                                                                                    |             |
| <b><u>Draw any two out of the following three questions.</u></b> |                                                                                                                                                                                                                                    |             |
| <b>Q. 2</b>                                                      | A line AB 70 mm long is inclined at 45° to HP and 30° to VP. End A is in 20 mm above HP and 30 mm in front of VP and the end B is in third quadrant. Draw the projections of the line using rotation method and locate its traces. | (05)        |
| <b>Q. 3</b>                                                      | The end projectors of a line CD are 60 mm apart. The end C is 30 mm above HP and 20 mm in front of VP. The end D is 20 mm above H.P. and 35 mm in front of VP. Determine the true length using auxiliary plane method.             | (05)        |
| <b>Q. 4</b>                                                      | A square ABCD of 50 mm side has its corner A in the H.P., its diagonal AC inclined at 30° to the H.P. and the diagonal BD inclined at 45° to the V.P. and parallel to the H.P. Draw its projections.                               | (05)        |

5.5

**Department of Mechanical Engineering, Punjabi University, Patiala**  
**MCE -151 Engineering Graphics, First Mid-Semester Exam, April-2022**  
**B. Tech. 1st Year, 1<sup>st</sup> Sem (Group B)**

**Max. Marks: 15**

**Time: 1 Hour**

**NOTE: Section A is compulsory. Attempt any two questions from Section B.**

| <b>Section A: Answer the following five questions with pen only.</b> |                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q.1</b>                                                           |                                                                                                                                                                                                                                                                                                                                            |
| (i)                                                                  | The front view of a point A is 20mm below XY line and top view is 35mm below XY line. The point A is 20mm _____ HP (Above / below) and 35mm _____ (in front of / behind) VP. 01                                                                                                                                                            |
| (ii)                                                                 | The apparent length of a line is always _____ (greater than / equal / less than) the true length of the straight line. 01                                                                                                                                                                                                                  |
| (iii)                                                                | A line is inclined to HP and parallel to VP. Its front view shows _____ and top view shows _____. (True length, Apparent Length, Point view) 01                                                                                                                                                                                            |
| (iv)                                                                 | The plane inclined to VP and perpendicular to HP is called _____. (Profile plane, AIP, AVP) 01                                                                                                                                                                                                                                             |
| (v)                                                                  | A rectangular lamina is parallel to HP and perpendicular to VP. Its true shape will appear on _____. (HP / VP / PP). 01                                                                                                                                                                                                                    |
| <b>Section B: Draw any two out of the following three questions.</b> |                                                                                                                                                                                                                                                                                                                                            |
| <b>Q. 2</b>                                                          | The distance between the end projectors of the line AB is 65 mm. The end A of the line is 45 mm above HP and 5 mm in front of VP and end B is 5 mm above HP and 40 mm in front of VP. Draw the projections of the line and find its true length, HT, VT and inclinations $\theta$ and $\phi$ using rotation method. 05                     |
| <b>Q. 3</b>                                                          | End A of the line AB is 45 mm above HP and 15 mm in front of VP. The distance between the end projectors is 75 mm. End B is 25 mm below HP and 50 mm behind VP. Draw the projections of the line and find its true length (in front view and top view). Also find its true inclinations $\theta$ & $\phi$ using auxiliary plane method. 05 |
| <b>Q. 4</b>                                                          | A regular pentagon of 35 mm side has one side on HP. Its plane is inclined at $30^\circ$ to HP and perpendicular to VP. The side on which it rests on HP is inclined at $45^\circ$ to VP. Draw the projections of the oblique lamina. 05                                                                                                   |